

Cryptocurrency Replication Using Machine Learning

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Abstract

Cryptocurrencies are characterized by high volatility and low correlations with traditional asset classes, and present an intriguing investment opportunity. However, their inherent risks and regulatory uncertainties make direct investment challenging for many investors. This paper addresses this challenge by proposing a replication framework that employs machine learning to create synthetic cryptocurrency portfolios that replicate the risk-adjusted return profile and diversification benefits of cryptocurrencies. We show that synthetic cryptocurrency portfolios offer a compelling alternative to direct investment in cryptocurrencies, delivering superior risk-adjusted returns net of trading costs while mitigating the risks that are associated with holding cryptocurrencies directly. Furthermore, synthetic cryptocurrency portfolios provide better diversification benefits and lower tail risk.

Keywords: Portfolio replication; Cryptocurrencies; Machine learning algorithms.

JEL codes: G10, G11

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Cryptocurrencies are characterized by high volatility and low correlations with traditional asset classes, and present an intriguing investment opportunity. However, their inherent risks and regulatory uncertainties make direct investment challenging for many investors. This paper addresses this challenge by proposing a replication framework that employs machine learning to create synthetic cryptocurrency portfolios that replicate the risk-adjusted return profile and diversification benefits of cryptocurrencies. We show that synthetic cryptocurrency portfolios offer a compelling alternative to direct investment in cryptocurrencies, delivering superior risk-adjusted returns net of trading costs while mitigating the risks that are associated with holding cryptocurrencies directly. Furthermore, synthetic cryptocurrency portfolios provide better diversification benefits and lower tail risk.

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1. Introduction

The last decade has witnessed very significant growth in the market capitalization of cryptocurrencies, from less than 100 million USD in 2012 to about two trillion USD in 2022 (Statista, 2022). During that time, cryptocurrencies have become increasingly important as a medium of exchange, a store of value and, most notably, a speculative investment asset. Indeed, cryptocurrencies are now considered an asset class in their own right and are widely held by both individual and institutional investors (see, for example, Corbet et al., 2019; Hajric, 2020). This rapid growth in cryptocurrencies has been closely followed by regulators, not least because it is well established that innovations in financial instruments can pose a threat to systemic stability (Minsky 1977; Beck et al., 2016). The regulatory response has varied considerably between countries, from banning cryptocurrencies altogether (such as in China and Russia) to endowing them with the status of legal tender (such as in El Salvador). More commonly, however, governments have chosen to permit the operation of cryptocurrencies, at least for now, but to simultaneously caution investors about the substantial risks arising from their unregulated nature and their high volatility, and to regulate the trading of cryptocurrency derivatives.¹ In the US, the Securities and Exchange Commission (SEC) has defined cryptocurrencies as securities and is expected to apply the corresponding regulations to digital wallets and exchanges, while the Commodities Futures Trading Commission (CFTC) has stated that it considers them as commodities and has allowed cryptocurrency derivatives to trade publicly. In contrast, in the United Kingdom, the Financial Conduct Authority (FCA) has banned the retail trading of derivatives and exchange-traded notes (ETNs) on cryptocurrencies, justifying this on the basis of

¹ For a detailed report of the global regulatory approach to cryptocurrencies, see US Congress (2018).

the significant volatility in the prices of cryptocurrencies, combined with the inherent difficulties in their valuation. Increasingly, the focus is shifting to the threat that cryptocurrencies pose to monetary policy: if they become widely accepted as a means of payment (in addition to being a speculative investment asset and a store of wealth) but remain outside of the regulated monetary system, the effectiveness of monetary policy tools is eroded. Indeed, foreseeing this possibility, a number of central banks have either introduced their own digital currency (such as China) or are considering this (such as Sweden and the UK). Such central bank digital currencies could be expected to compete with established cryptocurrencies, representing an additional risk for investors.

In this paper, we investigate whether it is possible to replicate the return distribution of cryptocurrencies by investing in conventional asset classes. The motivation for this is twofold. First, from an investor's point of view, there are very evident non-market risks associated with investing in cryptocurrencies arising from the unregulated nature of the market and of institutions operating within that market. These include operational risks such as technological failure, security breaches and fraudulent activities targeting cryptocurrencies; liquidity risks arising from low trading volumes, limited market depth and market manipulation; and legal risks arising from changes in regulation and government policy that may restrict or prohibit the ownership, trading, or exchange of cryptocurrencies, as well as potential regulatory uncertainties surrounding the legal status and treatment of digital assets. A naturally interesting question, therefore, is whether investors are able to access the risk-return trade-off that is offered by cryptocurrencies by investing in conventional financial assets that are highly liquid and properly regulated. Second, the unusual statistical properties of cryptocurrency returns offer a unique opportunity to test and further develop methods that are commonly used in constructing passively managed portfolios that are

able to replicate the return distributions of other alternative investments such as hedge funds. Cryptocurrency returns are characterized by extremely high volatility, fat tails and relatively low (and time-varying) correlations with other financial instruments, even those that are traditionally thought of as being safe haven assets, such as precious metals (see, for example, Liu and Tsyvinksi, 2018; Chen et al. 2016; Guesmi et al. 2019), and so their replication through investment in conventional asset classes represents a challenge for existing methods of portfolio replication, and an interesting opportunity to develop new methods that are better suited to the unique characteristics of cryptocurrencies.

There are two main approaches to the replication of the returns of a target portfolio: the factor-based approach and the distribution-matching approach. The factor-based approach minimizes the tracking error between the target portfolio return and the return of a replicating portfolio that invests in conventional assets using linear regression or nonlinear optimization (see, for example, Jaeger and Wagner, 2005; Hasanhodzic and Lo, 2007, Amenc et al., 2010). The distribution-matching approach matches the first two moments of the unconditional distribution of the target portfolio return with those of the replicating portfolio (see, for example, Kat and Palaro, 2005, 2006; Papageorgiou et al., 2008; Takahashi and Yamamoto, 2010).

We employ the distribution-matching approach using machine learning to accommodate the specific distributional characteristics of cryptocurrency returns and establish the relationship between cryptocurrency returns and a set of predefined factors or features. Specifically, we replicate the monthly returns of the most widely traded cryptocurrency, namely Bitcoin, by constructing leveraged long-only or long-short portfolios of conventional assets using various machine learning algorithms. Our investment universe comprises international equity, fixed income and alternative asset indices, all of which can be traded using highly liquid, investible

instruments such as exchange-traded funds, futures and options. Using out-of-sample evaluation, we show that the machine learning distribution-matching approach yields replicating portfolios that mimic the risk-adjusted performance and the distributional characteristics of the underlying cryptocurrency that they are designed to track. Our results are robust to realistic transaction costs, different estimation window lengths, and alternative short sale and leverage constraints.

A growing literature investigates whether cryptocurrencies can provide diversification benefits through their low correlation with other asset classes, and hence potentially enhance the risk-adjusted return performance of traditional portfolios. For example, Baur et al. (2018) investigate the diversification benefits of Bitcoin and suggest that it can enhance portfolio performance, especially for investors with a higher risk appetite. Similarly, Kajtani and Moro (2020) investigate the role of Bitcoin in well-diversified portfolios and find that its inclusion in a portfolio can enhance performance and provide diversification benefits in certain countries, particularly those with less developed financial markets. Chuen et al. (2017) find that the low but positive correlation of cryptocurrencies with traditional asset classes leads to some potential diversification; however, they emphasize that the unique regulatory and liquidity risks associated with cryptocurrencies may limit their potential benefits in practice. In contrast, Bouri et al. (2019) study the properties of Bitcoin in traditional portfolios but find limited evidence to support its role as a hedge or safe haven asset, suggesting that its contribution to portfolio diversification may be marginal. We provide further evidence on the diversification benefits of cryptocurrencies for investors, and test whether these benefits can be obtained through investment in the replicating portfolios that we obtain using the machine learning distribution-matching approach described above.

The outline of the remainder of the paper is as follows. Section 2 describes the data, portfolio replication framework, estimation methods and evaluation criteria. Section 3 reports the empirical

results. Section 4 provides a summary and some concluding remarks.

2. Data and Methodology

2.1 Data

The total market capitalization of cryptocurrencies is USD 797.05 billion as of 30.12.2022 down from the historic high of USD 2,967.48 billion on 10.11.2021. Of the 7,396 cryptocurrencies in existence, Bitcoin has by far the largest market share (40.2%), followed by Ethereum (18.4%) and Tether² (8.3%).³ As well as being the largest cryptocurrency by market capitalization, Bitcoin also has a much longer price history at 150 months, compared to 59 months for Ethereum and 65 months for Tether. Bitcoin has a very high correlation with other cryptocurrencies (for example, the correlation between Bitcoin and Ethereum prices over the period January 2018 to December 2022 is 0.92). For these two reasons, our analysis focuses only on Bitcoin. We obtained monthly Bitcoin closing prices from Bloomberg. The full sample covers the period January 2012 to December 2022, corresponding to 132 monthly observations.⁴

We consider an investment universe of 20 equity, fixed-income and alternative assets (including commodities and private equity), which we also obtained from Bloomberg. These indices represent the assets available to a typical international investor and can be traded via a range of low-cost, highly liquid instruments such as exchange-traded funds, futures and options. The first group of assets are US and international equity indices: Russell 1000 Growth (RU10GRTR), Russell 1000 Value (RU10VATR), Wilshire 5000 Total Market (W5000FTR),

² Tether (USDT) is a stablecoin, whose value is pegged to the US dollar.

³ www.coinmarketcap.com

⁴ Although Bitcoin started trading in July 2010, the first 12-18 months were characterised by low volume and extreme price movements. Trading volume increased significantly in 2012, and regulatory attention (in the form of published guidance) started in 2013.

Dow Jones Industrial (DJITR), MSCI Pacific (NDDUP), MSCI Europe (NDDUE15), MSCI World IT (MXWO0IT) and MSCI World Energy Stock (MXWO0EN). The second group comprises US and international fixed-income indices: US Corporate Bonds (LUACTRUU), US Aggregate Bond Market (EMUSTRUU), EU Aggregate Bond Market (LECP TREU) and German Sovereign Bonds (I14JD). The third group are alternative assets, including S&P GSCI Commodities (SPGSESTR), S&P GS Gold (SPGSGCTR), S&P GS Silver (SPGCSITR), S&P GS Platinum (SPGCPLTR), US Real Estate (FNRETR), US Crude Oil (USCRWTIC) and US Private Equity (LPX50TR). In the following, we refer to these three groups of indices as the replicating assets. Table 1 reports descriptive statistics for the monthly returns of Bitcoin and the replicating

Asset Class	Assets	Tickers	MEAN	MED	STD	MIN	MAX	SKEW	XKURT
Crypto Currency	Bitcoin	BTCUSD	0.068	0.054	0.238	-0.411	0.726	0.545	0.152
Equity	Pacific Stock	NDDUP	0.004	0.007	0.043	-0.114	0.132	-0.224	1.214
	Europe Stock	NDDUE15	0.004	0.005	0.049	-0.144	0.170	-0.061	0.930
	Emerging Markets	NDUEEGF	0.003	0.003	0.051	-0.154	0.148	0.034	0.694
	World IT	MXWO0IT	0.012	0.018	0.057	-0.120	0.135	-0.252	-0.400
	World Energy	MXWO0EN	0.004	0.004	0.079	-0.296	0.282	0.032	2.492
	US Large Cap Growth	RU10GRTR	0.011	0.014	0.051	-0.121	0.148	-0.226	0.156
	US Large Cap Value	RU10VATR	0.007	0.008	0.047	-0.171	0.135	-0.511	1.836
	US Stock Market	W5000FTR	0.009	0.014	0.047	-0.136	0.133	-0.369	0.680
	US Large Cap	DJITR	0.010	0.010	0.047	-0.136	0.141	-0.141	0.807
Fixed Income	US Corporate Bonds	LUACTRUU	0.002	0.003	0.019	-0.071	0.052	-0.648	2.513
	US Agg Bond Market	EMUSTRUU	0.002	0.004	0.022	-0.107	0.066	-1.397	6.514
	EU Agg Bond Market	LECP TREU	0.000	0.000	0.015	-0.069	0.047	-1.089	5.931
	German Sovereign	I14JD	0.004	0.012	0.061	-0.277	0.107	-1.161	3.415
Alternatives	Commodities	SPGSESTR	0.005	0.000	0.039	-0.080	0.106	0.260	-0.256
	Gold	SPGSGCTR	0.006	-0.004	0.079	-0.178	0.300	0.856	1.504
	Silver	SPGCSITR	0.001	-0.003	0.065	-0.158	0.139	-0.060	-0.519
	Platinum	SPGCPLTR	-0.002	-0.002	0.022	-0.063	0.051	-0.280	0.601
	US Real Estate	FNRETR	0.005	0.008	0.052	-0.219	0.118	-0.796	2.515
	US Crude Oil	USCRWTIC	0.014	0.021	0.144	-0.542	0.884	1.697	14.381
	US Private Equity	LPX50TR	0.009	0.016	0.066	-0.260	0.184	-0.706	2.424

Table 1: Descriptive Statistics of Cryptocurrency and Replicating Asset Returns. The table reports descriptive statistics for the cryptocurrency (Bitcoin) and the replicating asset returns over the Jan 2015-December 2022 period (96 months). The statistics include average (MEAN), median (MED), standard deviation (STD), minimum (MIN), maximum (MAX), skewness (SKEW) and excess kurtosis over normal distribution (XKURT). Replicating assets include nine equity (five international and four US equity) indices, four bond indices (two US and two international bond market indices) and seven alternative assets (commodities, real estate and private equity indices). The data source is Bloomberg.

assets over the full sample. Bitcoin returns have a significantly higher mean return and standard deviation than the returns of the other assets. They are highly non-normal, with very significant positive skewness and relatively high excess kurtosis. In contrast, the returns of the replicating assets have much lower mean returns and standard deviations, and lower (and often negative) skewness.

Table 2 reports the correlations between Bitcoin and each replicating asset. The US Private Equity, EU Bond Market and US Large Cap Growth indices have the highest correlation with Bitcoin. Notably, the correlation between Bitcoin and the replicating assets is positive in all cases, although close to zero for gold and crude oil. Table 2 also reports the time-series properties of the

Asset Class	Assets	CORR	ARCHtest	pval	Qstats	pval	JBstat	pval
Crypto Currency	Bitcoin	1	5.038	0.284	8.684	0.730	4.851	0.061
Equity	Pacific Stock	0.280	3.336	0.503	6.244	0.903	6.701	0.034
	Europe Stock	0.289	2.175	0.704	5.474	0.940	3.517	0.107
	Emerging Markets	0.208	1.681	0.794	6.346	0.898	1.944	0.288
	World IT Stock	0.318	12.104	0.017	9.825	0.631	1.653	0.349
	World Energy Stock	0.207	2.404	0.662	14.117	0.293	24.865	0.002
	US Large Cap Growth	0.329	11.548	0.021	12.332	0.419	0.915	0.500
	US Large Cap Value	0.297	14.023	0.007	17.730	0.124	17.669	0.005
	US Stock Market	0.325	12.468	0.014	13.002	0.369	4.022	0.085
	US Large Cap	0.317	12.138	0.016	16.481	0.170	2.924	0.149
Fixed Income	US Corporate Bonds	0.253	7.855	0.097	8.905	0.711	31.991	0.001
	US Agg Bond Market	0.254	4.935	0.294	17.371	0.136	200.924	0.001
	EU Agg Bond Market	0.338	12.879	0.012	7.947	0.789	159.663	0.001
	German Sovereign	0.191	1.990	0.738	10.618	0.562	2.700	0.171
Alternatives	Commodities	0.146	3.391	0.495	10.070	0.610	68.216	0.001
	Gold	0.067	2.347	0.672	5.088	0.955	1.341	0.427
	Silver	0.151	26.698	0.000	7.440	0.827	20.755	0.003
	Platinum	0.231	3.607	0.462	11.164	0.515	1.132	0.500
	US Real Estate	0.248	2.946	0.567	6.922	0.863	35.445	0.001
	US Crude Oil	0.079	9.501	0.050	13.590	0.328	873.329	0.001
	US Private Equity	0.341	6.360	0.174	7.768	0.803	31.467	0.001

Table 2: Time Series Properties of Cryptocurrency and Replicating Asset Returns. The table reports the correlation coefficients between the cryptocurrency and the replicating assets (CORR), Engle (1982) 's ARCH test statistics (ARCH) and the p -value of the test statistics up to 4-lags, Ljung-Box portmanteau test for serial correlation (Qstats) test statistics and the p -value of the test statistics calculated up to 12-lags, Jarque-Bera normality test statistics (JBstat) and the p -value of the test statistics (pval).

returns of Bitcoin and the replicating assets. In particular, it gives the results of the Ljung-Box test for serial correlation up to 12 lags and the ARCH test of Engle (1982) for up to four lags. The ARCH test shows no evidence of volatility clustering in monthly Bitcoin returns, and the same is true for all of the replicating assets except the silver and US Large Cap Value indices. The Jarque–Bera test rejects the null hypothesis of normality for Bitcoin and the majority of equity indices, but fails to reject normality for most of the fixed income and alternative assets.

The statistical analysis reported in Tables 1 and 2 thus suggests that Bitcoin returns differ markedly from those of established financial instruments, highlighting its low correlations with other assets, higher average returns, higher volatility and significant non-normality. We aim to reproduce these characteristics through investment in a portfolio of the replicating assets.

2.2 Replication Framework

Existing frameworks for replicating asset returns (most notably those of hedge funds) tend to focus only on the first and second moments of the return distribution. However, it is clear from the preceding discussion that Bitcoin returns are highly non-normal and so accommodating this feature in the replication framework is likely to add significant explanatory power. We employ commonly used frameworks mainly developed for hedge fund return replication as the benchmark approaches, and propose four machine learning approaches to capture the specific distributional characteristics of cryptocurrency returns.

2.2.1 Benchmark Models: Factor-Based Approach

The first set of benchmarks we use employ the well-established factor-based approach, both in its most basic form and supplemented with constraints on the mean return and downside risk of the replicating portfolio. We first define the standard factor-based model that minimizes the tracking error between Bitcoin and the replicating portfolio with no reference to the return

distribution. Specifically, defining the returns of the replicating assets $i = 1, \dots, N$ at time t as $r_{i,t}$, and the replicating portfolio weights as x_i , the objective function for the standard factor-based model (F1) is given by:

$$\min_x f(x) := \text{var}(r_{p,t} - r_{B,t}) \quad (1)$$

subject to the following budget and asset constraints

$$\sum_{i=1}^N x_i \leq L \quad (2)$$

$$0 \leq x_i \leq 1, i = 1, \dots, N \quad (3)$$

where $r_{p,t} = \sum_{i=1}^N x_i r_{i,t}$ is the replicating portfolio return at time t , $r_{B,t}$ is the Bitcoin return at time t and $\text{var}(\cdot)$ is the sample variance. The budget constraint (2) allows a maximum total leverage L . Constraint (3) requires the replicating portfolio to hold long-only positions in the replicating assets and restricts the maximum weight in any individual asset to 100 percent. The maximum and minimum weights for the long-short replication portfolio are 100 percent and -100 percent, respectively. The second factor-based model (F2) additionally imposes an equality constraint on the mean return:

$$\mu_p = \mu_B \quad (4)$$

where μ_p and μ_B are the sample means of $r_{p,t}$ and $r_{B,t}$, respectively. The third factor-based model supplements the optimization problem given by (1)-(4) with a constraint that matches the downside risk of Bitcoin with that of the replicating portfolio. This accommodates the significant non-normality in Bitcoin returns since downside risk is a function of the higher moments of the return distribution as well as its mean and variance. We consider the Conditional Value-at-Risk (CVaR) constraint:

$$CVaR_p^\alpha = CVaR_B^\alpha \quad (5)$$

where $CVaR_p^\alpha$ and $CVaR_B^\alpha$ are the α -percent CVaR of the replicating portfolio and Bitcoin, respectively. The replication model F3, corresponding to (1)-(5), is the hybrid approach proposed by Harris and Mazibas (2012) for the replication of hedge fund returns.

2.2.2 Benchmark Models: Distribution-Matching Approach

The second approach to replicating returns is the distribution-matching approach, commonly used in the analysis of hedge fund replication. This involves constructing a replicating portfolio that is designed to mimic the return distribution of a target hedge fund or a hedge fund index. The approach aims to capture the statistical distribution of hedge fund returns without necessarily tracking their period-by-period returns. Specifically, the objective function of the standard distribution-matching model (M1) is given by:

$$\min_x f(x) := |\mu_p - \mu_B| + |\sigma_p - \sigma_B| \quad (6)$$

subject to the budget and asset constraints:

$$\sum_{i=1}^N x_i \leq L \quad (7)$$

$$0 \leq x_i \leq 1, i = 1, \dots, N \quad (8)$$

where σ_p and σ_B are the sample standard deviations of $r_{p,t}$ and $r_{B,t}$, respectively.

2.2.3 Machine Learning Approaches: Regression Trees

Regression trees are a type of decision tree model used for regression tasks and are able to capture nonlinear relationships between the input features and a continuous target variable. The training process of a regression tree involves recursively partitioning the feature space based on certain criteria to create a hierarchical structure. At each step, the algorithm selects a feature and a

split point that optimally divides the data into two subsets, maximizing the homogeneity of the target variable within each subset. This process continues until a stopping criterion is met, such as reaching a maximum depth or a minimum number of samples in each leaf node. Once the regression tree is trained, it can be used to make predictions by traversing the tree from the root node to a leaf node based on the values of the input features. The predicted value at the leaf node is then assigned as the output. We train the regression tree model (model ML1) using historical data on the replicating asset returns (the input variables) and the corresponding Bitcoin returns (the target variable). After training the model, we use it to predict Bitcoin returns and extract the portfolio weights from the model, which we use to construct the replicating portfolio.

2.2.4 Machine Learning Approaches: Ensemble Regression Trees

Ensemble regression trees (ERT) combine multiple individual regression trees to create a more accurate and robust predictive model for regression problems. In an ensemble regression tree, multiple regression trees are trained on different subsets of the training data and their predictions are aggregated to yield a final single prediction. We implement the ERT model (model ML2) using the Random Forest approach, which builds an ensemble of regression trees by randomly selecting a subset of features (the input variables) at each split. We train each tree independently on a different subset of the training data, using bootstrap aggregating (bagging) and random forest learning. We use 5% k -fold cross-validation and optimize the number of learning cycles (in the range [10,500]) and learning rate (in the range [0.0001,1]) using Bayesian optimization and the default tree structure.

2.2.5 Machine Learning Approaches: Support Vector Machines

Unlike traditional regression methods that minimize errors between the data and the fitted hyperplane aggregated across all data points, Support Vector Machine (SVM) regression finds the

hyperplane that maximizes the margin between the hyperplane and support vectors, which are the data points that are closest to the hyperplane (Vapnik et al. 1995; Vapnik et al. 1997). Using the kernel trick, SVM Regression are able to handle nonlinear relationships between input features and the target variable.⁵ In particular, it allows SVM Regression to implicitly map the input features to a high-dimensional feature space, where the data might be linearly separable.⁶ SVM Regression incorporates a regularization parameter to control the trade-off between model complexity and the deviation allowed in the training data, and a parameter that determines the width of the insensitive tube around the hyperplane. Data points falling within this ε -tube are considered to have acceptable deviations. The objective is to minimize the regularized empirical risk function while maintaining the maximum margin and ensuring deviations are within the ε -tube. The solution involves finding the optimal values of the model's weights and biases (Smola and Schölkopf, 2004). The training of SVMs is equivalent to solving a linearly constrained quadratic programming problem and hence the solution is a unique global minimum, in contrast with traditional neural network models which require nonlinear optimization (see Tay and Cao, 2001). We train the SVM regression (model ML3) to find the most accurate ensemble trees using 5% k -fold cross-validation, and optimize the following hyperparameters: box constraint for alpha coefficients (in the range [0.001,1000], kernel scale parameter (in the range [0.001,1000], and the tube size (in the range $[0.001,100] * iqr(r_{B,t})/1.349$, where $iqr(.)$ is the interquartile range) using Bayesian optimization. We use a linear kernel function and polynomial order of three.

⁵ Using the kernel trick, SVMs can efficiently perform nonlinear classification or regression by implicitly mapping the data into a higher-dimensional feature space without explicitly computing the transformed feature vectors. See Schölkopf and Smola (2002) for details.

⁶ Popular kernel functions include linear, polynomial, radial basis and sigmoid.

2.2.6 Machine Learning Approaches: Artificial Neural Networks

Artificial Neural Network (ANN) regression consists of interconnected layers of artificial neurons, including an input layer, one or more hidden layers, and an output layer. Neurons in each layer receive input signals, apply a nonlinear activation function to compute their output and pass it to the next layer. The weights and biases of the neurons are iteratively adjusted during the training process to optimize the model's performance. Activation functions introduce nonlinearity to the neural network enabling it to capture complex patterns in the data. Popular activation functions include the sigmoid, hyperbolic tangent (tanh), and rectified linear unit (ReLU) functions (see, for example, Goodfellow et al., 2016). ANN regression employs backpropagation proposed by Rumelgart et al. (1986), which is a learning algorithm that adjusts the weights and biases based on the difference between the predicted and actual target values. Gradient descent is commonly used to update the model parameters by iteratively minimizing the mean squared error or other loss functions. The process involves calculating the gradients, propagating them backwards through the network, and updating the weights and biases. ANN regression is prone to overfitting, where the model becomes too complex and performs well on the training data but poorly on unseen data. Regularization techniques such as L1 and L2 regularization, dropout, and early stopping can be used to mitigate overfitting and improve generalization (Bishop, 1995).

We employ a feedforward neural network (model ML4) with fully connected layers for regression analysis. The neural network architecture comprises multiple layers, each connected to the preceding layer. The initial layer receives input data, representing the predictor variables. The input is multiplied by a weight matrix in each subsequent layer and augmented with a bias vector. Following each fully connected layer, an activation function is applied, except for the final layer. The last fully connected layer produces the network's output, providing predicted response values.

We train the regression ANN with four layers of sizes 3, 5, 10 and 250, respectively, with no hyperparameter optimization. We use the rectified linear unit (ReLU) activation function for fully connected layers, zero value for the regularization term strength, zero value for initial fully connected layer biases and Glorot (also known as the Xavier initializer) as the function to initialize fully connected layer weights.

2.3 Estimation Methodology

We test the out-of-sample performance of the replicating portfolios from January 2018 to December 2022, which captures a complete market cycle for Bitcoin. We consider the following benchmark portfolios: (a) the factor-based model minimizing tracking error (F1); (b) the factor-based model minimizing tracking error with mean constraint (F2); (c) the factor-based model with mean and CVaR constraint (F3); (d) the distribution-matching model matching mean and variance (M1); and the following machine learning-based portfolios: (a) regression tree (ML1), (b) ensemble regression tree (ML2); (c) support vector machine (ML3); (d) artificial neural network (ML4) and (e) an equally weighted combination of these four machine learning portfolios (COMB). We consider two versions of each portfolio: a long-only (LO) version that restricts the asset weights to be non-negative, and a long-short (LS) version that allows short selling.

We initially estimate the replication models using the first 36 months (January 2015 to December 2017) to make out-of-sample forecasts of the replicating portfolio weights for January 2018. We then roll the estimation sample forward one month (February 2015 to January 2018) to forecast the portfolio weights for February 2018, and so on until the end of the sample is reached in December 2022. We hold the replicating portfolio for one month and then rebalance it. The estimation window length is kept constant at 36 months. The optimization problems in (1)-(6) are solved using the Matlab *fmincon* nonlinear solver with the *GlobalSearch* algorithm. In estimating

the CVaR constraint, we use a 99 per cent confidence level. Matlab's Machine Learning Toolbox is used to estimate the machine learning algorithms.

We use the approaches described above to rebalance the replicating portfolios each month. The replicating portfolios exclusively use exchange-traded funds (ETF) or derivatives such as futures on the indices listed in Table 1. It is assumed that the investor holds a margin account with a broker. The results reported in the main analysis in Section 3.1 do not employ leverage. As a robustness check, we also consider the case where the investor is able to take positions up to 20 times the margin account balance. If the total exposure exceeds the maximum allowed leverage in any month, the latter is reduced to the permitted level by selling assets. If a model suggests a reduction in leverage, the investor reduces the leverage by liquidating the positions and paying back the borrowed funds. We assume that the investor pays interest of LIBOR+50bp for borrowed funds and also pays a 10bp fee for all (buy and sell) trades. The investor can trade as long as there is a positive balance in the margin account and the actual leverage is lower than the maximum allowed total leverage. The account is suspended whenever the margin account balance becomes zero or negative, and positions are liquidated to repay the borrowed funds.

We evaluate the performance of the replicating portfolios using a range of metrics: return (i.e., Cumulative Average Growth Rate, CAGR), moments (i.e., standard deviation, skewness, kurtosis), risk-adjusted return (i.e., Sharpe ratio, SR), risk (i.e., Maximum Drawdown, MDD), tracking error (TE), a statistical distance measure (i.e., minimum absolute error distance, D_{mae}) and portfolio turnover (PT). We also assess the explanatory power of each replicating portfolio by regressing the returns of Bitcoin on the replicating portfolio returns over the out-of-sample period and report the regression R^2 statistics.

2.4 Measuring Diversification Benefits

As noted in the introduction, the potential diversification benefits of cryptocurrencies for existing investment portfolios have been evaluated in a number of studies. We explore whether the replicating portfolios that we construct are able to offer similar diversification benefits for a typical international investor. We use a representative replicating portfolio generated by the machine learning model ML3, and separately consider both the long-only and long-short versions. We analyse the case of a '60/40' investor who invests 60 percent of her wealth in equity and 40 percent in fixed income. The investor uses either minimum-variance or mean-variance optimization to obtain the optimal weights within the equity and fixed income asset classes. We assume that the investor invests 10 percent of her portfolio in either Bitcoin or the replicating portfolio to yield a 55/35/10 portfolio, and we evaluate the portfolio effects in each case. For the allocation of assets within the equity and fixed income asset classes, we consider the following optimization approaches: the global minimum variance portfolio (GMV), the 1-norm constrained minimum variance portfolio strategies of DeMiguel et al. (2009), calibrated using cross-validation over portfolio variance (NC1V), the maximum Sharpe ratio portfolio (SR), and the Bayesian Shrinkage portfolio of Ledoit and Wolf (2003, 2004) (BS). We first evaluate the contributions of Bitcoin and the replicating portfolio in terms of changes in the portfolio CGAR, SR, MDD, CVaR, turnover, skewness and kurtosis. We then apply spanning tests for the portfolios separately constructed with Bitcoin and the replicating portfolio. For the spanning tests, we consider all of the replicating portfolios described above. In the robustness analysis below, we also consider the case of an investor who is not constrained to the 60/40 allocation, and can additionally invest in alternative assets. For both the 60/40 and unconstrained cases, we assume that investors construct

their core portfolios using a broader range of assets than the replicating assets defined in Section 2.1.⁷

3. Empirical Results

3.1 Replication Performance

In this section, we compare the out-of-sample replication performance of the four benchmark portfolios (F1, F2, F3 and M1) and the five machine learning-based portfolios (ML1, ML2, ML3, ML4 and COMB) over the period January 2018-December 2022. Here, we provide the results for non-leveraged portfolios only. To explore the effect of leverage on the replication performance, we consider a range of leverage levels in the robustness analysis below. The out-of-sample replication results are reported in Table 3, together with the corresponding results for Bitcoin. Results for the long-only (LO) replicating portfolios are reported in Panel A and for the long-short (LS) replicating portfolios in Panel B.

⁷ The assets available to the investors are: (1) *World Stock*: MSCI World, MSCI Pacific, MSCI Europe, MSCI Emerging Markets, MSCI World IT, MSCI World Energy indices; (2) *US Stock*: US stock market (Wilshire 5000 Total Market), US large cap (Dow Jones Industrials) US large cap growth (Russell 1000 Growth), US large cap value (Russell 1000 Value), US small cap growth (Russell 2000 Growth), US small cap value (Russell 2000 Value), US Largest 3000 growth (Russell 3000 Growth), US Largest 3000 value (Russell 3000 Value), US midcap growth (Russell Midcap Growth), US midcap value (Russell Midcap Value); (3) *Fixed Income market* indices: US corporate bond market (Bloomberg Barclays US Corporate Bond), US aggregate bond market (Bloomberg Barclays US Aggregate), US credit market (Bloomberg Barclays US Credit), emerging markets aggregate (Bloomberg Barclays EM Aggregate), Europe aggregate (Bloomberg Barclays Euro-Aggregate), Europe corporate bond (Bloomberg Barclays Euro Aggregate Corporate), Europe aggregate treasury (Bloomberg Barclays Euro Aggregate Treasury), Sovereign Germany market (Markit iBoxx Sovereign Germany 1-10 TR), Japanese Government (Japan 10 year Government Bond Index) and UK Government (UK 10 year Gilt Bond Index); (4) *Alternatives*: gold (S&P GSCI Gold), silver (S&P GSCI Silver), platinum (S&P GSCI Platinum), precious metals (S&P GSCI Precious Metals), energy (S&P GSCI Energy), real estate (FTSE Nareit US Real Estate Index), Oil (US Crude Oil WTI Index), private equity (LPX50 Listed Private Equity Index TR).

Models/ Metrics	CAGR	STD	SKW	KURT	DMAE	SR	TE	R ²	MDD	PT
BTC	0.033	0.809	0.378	2.526		0.115			0.757	
Panel A: Long-Only Replication Portfolios										
F1	-0.008	0.112	-0.654	4.641	0.181	-0.051	0.784	0.083	0.272	0.393
F2	-0.006	0.119	-0.769	3.909	0.179	-0.057	0.778	0.108	0.286	0.328
F3	0.004	0.107	-0.382	3.649	0.181	-0.013	0.787	0.072	0.246	0.482
M1	0.009	0.141	-0.491	3.100	0.174	-0.020	0.766	0.145	0.269	0.616
ML1	0.040	0.170	-1.126	6.055	0.173	0.043	0.761	0.144	0.254	0.371
ML2	-0.025	0.185	-1.121	6.568	0.176	-0.063	0.766	0.117	0.267	1.288
ML3	0.058	0.145	-0.671	3.490	0.176	0.089	0.773	0.109	0.223	0.396
ML4	0.081	0.170	0.354	5.121	0.176	0.103	0.773	0.098	0.200	1.217
COMB	0.029	0.147	-0.828	3.753	0.175	0.028	0.767	0.134	0.238	0.363
Panel B: Long-Short Replication Portfolios										
F1	-0.078	0.095	-1.235	11.859	0.194	-0.313	0.824	0.012	0.361	0.502
F2	-0.063	0.100	-1.362	11.403	0.192	-0.257	0.821	0.004	0.336	0.493
F3	-0.074	0.100	-1.578	11.307	0.194	-0.289	0.826	0.012	0.369	0.348
M1	-0.054	0.056	0.414	4.739	0.187	-0.382	0.806	0.006	0.253	1.400
ML1	0.040	0.170	-1.125	6.055	0.173	0.043	0.761	0.144	0.254	0.371
ML2	-0.016	0.131	-0.405	4.905	0.181	-0.060	0.792	0.043	0.272	1.397
ML3	0.164	0.155	-0.218	4.396	0.190	0.241	0.829	0.001	0.185	0.647
ML4	-0.105	0.162	-3.080	18.202	0.197	-0.229	0.850	0.026	0.500	1.374
COMB	0.125	0.145	0.426	3.089	0.192	0.199	0.824	0.000	0.166	0.557

Table 3: Out-of-Sample Replication Performance. The table reports the out-of-sample performance statistics for the benchmark and proposed replication portfolios: Cumulative Average Growth Rate (CAGR), standard deviation (STD), skewness (SKW), kurtosis (KURT), minimum absolute error distance measure (DMAE), Sharpe ratio (SR), tracking error statistics (TE), regression R² statistics of the cryptocurrency returns regressed on the replicating portfolios' out-of-sample returns (R²), maximum drawdown (MDD) and portfolio turnover (PT). Benchmark portfolios are (1) standard factor model minimizing tracking error (F1), (2) factor model minimizing tracking error with mean return constraint (F2), (3) factor model minimizing tracking error with downside risk (CVaR) constraint (F3) and (4) distribution matching model that matches portfolio mean and variance (M1). Proposed machine learning models: (a) Regression Trees (ML1), (b) Ensemble Regression Trees (ML2); (c) Support Vector Machines (ML3); (d) Artificial Neural Networks (ML4) and (e) COMB is an equally weighted combination of the proposed synthetic (replication) portfolios. Estimation sample size is 36 months. The reported results for the replicating portfolios do NOT use leverage. The results with several leverage levels are discussed in the Robustness Analysis section. The portfolio results are net of all financing and transaction costs. The investor is assumed to pay an interest of LIBOR+50bp for borrowed funds for long positions and LIBOR+100bp for borrowed funds or securities for short positions. The investor also pays 10bp trading expenses for all buy and sell trades. The portfolio results are net of all financing and transaction costs.

The four benchmark models all underperform in terms of CAGR relative to Bitcoin. The machine-learning models generally outperform the benchmark models, generating CAGRs closer to Bitcoin except for ML2 (both LO and LS) and ML4 (LS). The machine learning models also outperform Bitcoin in all instances, with the best performance generated by ML4 among the LO portfolios and ML3 among the LS portfolios. The standard deviation of Bitcoin is 80.9%, which

is between four and seven times higher than that of any of the replicating portfolios. The benchmark models have the lowest standard deviations, with F2 the lowest of the LO portfolios and M1 the lowest of the LS portfolios. Owing to the characteristics of the underlying asset returns, most of the replicating portfolios have difficulty in generating the skewness of Bitcoin (0.378), although ML4 achieves the closest skewness level among the LO portfolios (0.354); among the LS portfolios, M1 and COMB portfolios exhibit even higher skewness. The replicating portfolios generally exhibit higher kurtosis than Bitcoin, with M1 being the closest among the LO portfolios and COMB the closest among the LS portfolios. In general, however, the replicating portfolios match the higher moments of Bitcoin reasonably well, with the machine learning models generating the closest skewness and kurtosis, as well as a higher CAGR. In terms of risk-adjusted return, the benchmark replication models are unable to generate a positive Sharpe ratio, but in contrast, with the exception of ML2, the machine learning models consistently generate not only a positive Sharpe ratio, but in some cases, such as ML3 and COMB for the LS portfolios, one that is higher than that of Bitcoin.

The tracking error statistic measures how closely the replication portfolios mimic the month-by-month performance of Bitcoin, as opposed to just matching the unconditional distribution of Bitcoin returns. The machine learning models generally outperform the benchmark models, and among them, ML1 exhibits the lowest tracking error for both the LO and LS portfolios. A similar pattern is observed using both the statistical distance measure, D_{mae} , and the regression R^2 , with the machine learning models generating replicating portfolio returns that have higher explanatory power for Bitcoin returns than the benchmark models, with ML1 providing the highest R^2 among all of the models for both the LO and LS portfolios.

Bitcoin exhibits a very high maximum drawdown (MDD) and all of the models generate LO and LS portfolios with significantly lower MDD, with the machine learning models having a lower MDD than the benchmark models. As discussed in the robustness analysis below, the high MDD of Bitcoin is only achievable with high levels of leverage. Finally, we evaluate the portfolio turnover (PT) associated with the replicating portfolios. Benchmark models generally exhibit lower PT than machine learning algorithms, with F1 having the lowest of the LO portfolios and F2 lowest of the LS portfolios, although ML1 has a similar PT for both LO and LS portfolios and COMB a similar PT for the LO portfolios.

3.2 Portfolio Effects

In this section, we evaluate the portfolio effects of making an additional investment in Bitcoin versus an investment in the replicating portfolio, represented by the ML3 portfolio. Table 4 presents the changes in portfolio metrics arising from diversification using either Bitcoin or the replicating portfolio over the out-of-sample period from January 2018 to December 2022. Panel A reports the results for the Sharpe ratio and CAGR. The inclusion of either Bitcoin or the replicating portfolio increases the Sharpe ratio of the BS and SR mean-variance strategies but significantly reduces the Sharpe ratios of the GMV and NC1V minimum-variance strategies. The use of the replicating portfolio generally leads to Sharpe ratios that are similar or higher relative to the inclusion of Bitcoin. The CAGR is increased in all cases, with the improvement being more significant for the two minimum-variance strategies. For all four strategies, the inclusion of Bitcoin leads to a larger increase in CAGR compared to the inclusion of the replicating portfolio.

Strategy	BITCOIN	LO SCP	LS SCP	BITCOIN	LO SCP	LS SCP	BITCOIN	LO SCP	LS SCP
Panel A:	Portfolio Allocation (%)			Δ% Sharpe ratio			Δ% CAGR		
GMV	10.00	10.00	10.00	-116.54	-46.33	-127.06	213.98	38.78	124.24
NC1V	10.00	10.00	10.00	-120.22	-49.82	-124.53	224.18	50.00	130.21
BS	10.00	10.00	10.00	1.95	1.09	23.14	28.93	0.12	14.73
SR	10.00	10.00	10.00	16.16	3.95	22.37	54.12	2.10	11.76
Panel B:	Δ% MDD			Δ% CVaR			Δ% Turnover		
GMV	48.68	-2.32	-3.72	76.12	-1.02	-3.46	25.67	-17.34	-2.92
NC1V	49.36	-2.71	-3.60	76.12	-2.06	-2.99	-9.95	-19.62	-1.57
BS	20.88	-2.01	-8.93	-2.40	9.35	-8.22	-17.35	-9.71	-9.41
SR	31.74	-2.76	-0.79	8.93	-2.20	1.43	-28.42	-36.50	-30.66
Panel C:	Δ% Standard Deviation			Δ% Skewness			Δ% Kurtosis		
GMV	48.68	2.06	-7.47	68.54	60.76	71.49	-13.71	-3.57	2.70
NC1V	49.36	2.06	-7.85	75.09	63.62	74.24	-14.48	-4.07	3.79
BS	20.88	-0.72	-6.47	-34.56	14.82	17.66	-15.62	3.86	9.07
SR	31.74	0.00	-6.77	-67.51	23.76	39.59	-0.90	1.46	7.13

Table 4: Contributions to Portfolio Performance: Portfolio Metrics. The table reports traditional 60 Equity and 40 Fixed-income (60/40) mean-variance portfolios. The portfolios are constructed by using the mean-variance portfolio strategies, including the Global Minimum Variance (GMV), 1-norm (NC1V) constrained minimum variance portfolio strategies of DeMiguel et al. (2009), maximum Sharpe ratio (SR), and Bayesian Shrinkage (BS) portfolio of Ledoit and Wolf (2003, 2004). The portfolios do not use leverage and are long-only. The models construct out-of-sample portfolios with and without Bitcoin and with and without long-only (LO) and long-short (LS) synthetic cryptocurrency portfolios (SCP) over the out-of-sample period. The portfolio allocations refer to the average allocations to Bitcoin and SCP over the out-of-sample period. The 60/40 portfolio is assumed to have a fixed 10 per cent allocation to Bitcoin or SCP. Panel A reports portfolio allocation and percentage change (Δ%) in portfolio CAGR and Sharpe ratios after including Bitcoin or SCP. The Δ% Sharpe ratio refers to the percentage change in the Sharpe ratio of the total portfolio after investing in Bitcoin or SCP. The Δ% CAGR refers to a percentage change in the cumulative average growth rates (CAGR) in the investor portfolios after investing in Bitcoin or SCP. Panel B reports percentage change in portfolio risk measures (maximum drawdown, MDD), conditional value-at-risk, CVaR) and portfolio turnover. Panel C reports the percentage change in the portfolio standard deviation, skewness and kurtosis of the investor portfolios. We use the out-of-sample ML3 replication portfolios for synthetic portfolios as detailed in sections 2.3 and 3.1. We consider the transaction costs as detailed in section 2.3 and 2.4. All portfolio analytics are net of transaction costs over the out-of-sample period.

Panel B of Table 4 reports changes in maximum drawdown (MDD), conditional value-at-risk (CVaR) and portfolio turnover. Notably, while the inclusion of Bitcoin significantly increases MDD for all four strategies, the use of the replicating portfolio leads to a small reduction in MDD. Similarly, investing in Bitcoin generally increases CVaR, except for the BS strategy, which shows a slight reduction. Investing in the replicating portfolio in most cases leads to a small reduction in CVaR. Portfolio turnover is reduced in all cases, except for Bitcoin in the GMV strategy.

Panel C reports the changes in the standard deviation, skewness and kurtosis. Investing in Bitcoin leads to a significant increase in portfolio standard deviation in all cases, and a reduction

in skewness for the BS and SR strategies but an increase in skewness for the GMV and NC1V strategies. For all four strategies, the addition of Bitcoin reduces kurtosis reflecting diversification in tail events. In contrast, investing in the replicating portfolio leads to either similar or lower standard deviation, a significant increase in skewness, and a small but variable impact on kurtosis suggesting that it offers superior risk diversification benefits overall relative to Bitcoin.

We also examine the similarity of portfolio returns with an investment in Bitcoin and the replicating portfolio. In particular, we employ a spanning test in which we regress the returns of the portfolio with an investment in the replicating portfolio on the returns of the portfolio with an investment in Bitcoin portfolio. We undertake the spanning test using all of the replicating portfolios considered in Section 3.1. Table 5 reports R^2 values of these spanning tests for the LO replicating portfolios (Panel A) and the LS replicating portfolios (Panel B). We conduct the spanning tests for the 60/40 portfolios constructed using the four optimization approaches described above. On average, portfolios diversified using the replicating portfolios explain about 75 percent of the variation in the return of portfolio that is diversified using Bitcoin. Moreover, the machine learning models consistently display higher R^2 values than the benchmark models.

Portfolio	F1	F2	F3	M1	ML1	ML2	ML3	ML4	COMB
Panel A: Long-Only Replication Portfolios									
GMV	70.52	71.12	70.03	72.52	72.66	72.05	71.87	71.15	72.38
MVU	98.36	98.40	98.36	98.46	98.48	98.45	98.43	98.43	98.45
SR	75.34	75.32	75.10	76.47	77.00	76.35	75.67	75.97	76.49
NC1V	70.66	71.08	70.08	72.52	72.71	72.25	71.81	71.23	72.64
Panel B: Long-Short Replication Portfolios									
GMV	68.26	68.21	67.90	70.41	72.98	71.86	66.88	65.53	64.93
BS	81.89	82.04	81.82	82.39	83.48	82.70	80.98	79.96	80.57
SR	73.94	73.69	73.48	75.26	75.74	74.98	71.51	73.52	72.62
NC1V	68.42	67.91	68.11	70.84	72.99	71.98	66.99	65.66	65.18

Table 5: Contributions to Portfolio Performance: Spanning Test R-square values. The table reports the R-square values of the spanning tests where the out-of-sample returns of the portfolios with an investment in long-only (Panel A) or long-short (Panel B) synthetic portfolios (SCP) are regressed on the returns of the portfolios with Bitcoin investment.

3.3 Robustness Analysis

To establish the robustness of the replication results reported in section 3.1, we considered four different estimation window sizes (i.e., 24, 36, 48 and 60 months) and four different levels of portfolio leverage (i.e., 3, 5, 10 and 20 times the invested capital) for the long-only and long-short replication portfolios over the out-of-sample period. Here, we summarise the findings of this analysis.⁸ For long-only replication portfolios, increasing the window size increases the Sharpe ratio and CAGR for the benchmark and the proposed portfolios. However, increasing the maximum allowed leverage levels leads to a reduction in the Sharpe ratio for all portfolios except ML4. Unlike the benchmark models, the machine learning models consistently outperform Bitcoin, with a higher CAGR over most estimation window sizes and leverage levels. However, increasing the window size does not lead to better performance for long-short portfolios: all four machine learning models achieve their optimal performance with a 36-month sample size. Increasing leverage levels leads to a consistent decline in CAGR for all models.

The impact of increasing the estimation window size on MDD performance is not as clear-cut as the impact of increasing leverage levels. Most models display lower MDD as the estimation window size increases. Overall, the level of leverage exacerbates portfolio drawdown for all portfolios except for those generated by the machine learning models, for which MDD only marginally increases, while increasing the window size mostly leads to lower MDD. It is only possible to achieve the very high drawdown levels displayed by Bitcoin at high leverage levels for the replicating portfolios. Increasing the window size generally tends to lead to higher R^2 values. However, increasing leverage tends to lead to lower R^2 values for most models. Increasing the window size tends to decrease portfolio turnover for most models.

⁸ Detailed results for the robustness analysis are available on request from the authors.

To establish the robustness of the portfolio contribution results reported in section 3.2, in addition to the 60/40 investor, we also considered an unconstrained investor (i.e. without strict asset class allocation constraints) who uses either minimum-variance or mean-variance optimization to obtain the optimal weights globally. The performance results are comparable to those for the 60/40 investor.

4. Conclusion

In this paper, we show how investors are able to achieve broadly the same risk and return profile of Bitcoin through investment in synthetic portfolios constructed from highly liquid, low-cost conventional assets. In so doing, investors are able to avoid the wide range of non-market risks that are associated with investment in cryptocurrencies. The construction of such synthetic portfolios is a challenging task owing to the extremely low exposure of Bitcoin to systematic factors, and traditional approaches to portfolio replication are generally unable to meet this challenge. We propose several machine learning replication models employing regression trees, ensemble trees, Support Vector Machines and Artificial Neural Networks. The machine learning replication models that we propose are able to broadly capture the complex statistical properties of Bitcoin returns and in many cases provide a similar, or indeed superior, return distribution. The replication approaches introduced in this paper could be extended beyond the cryptocurrency space. The proposed frameworks can easily be used to reproduce the statistical properties of any financial or non-financial asset returns using highly liquid, widely accessible and cheap-to-trade financial instruments. The models can also be used for constructing passive portfolios that aim to track the performance of benchmark market indexes.

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